

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

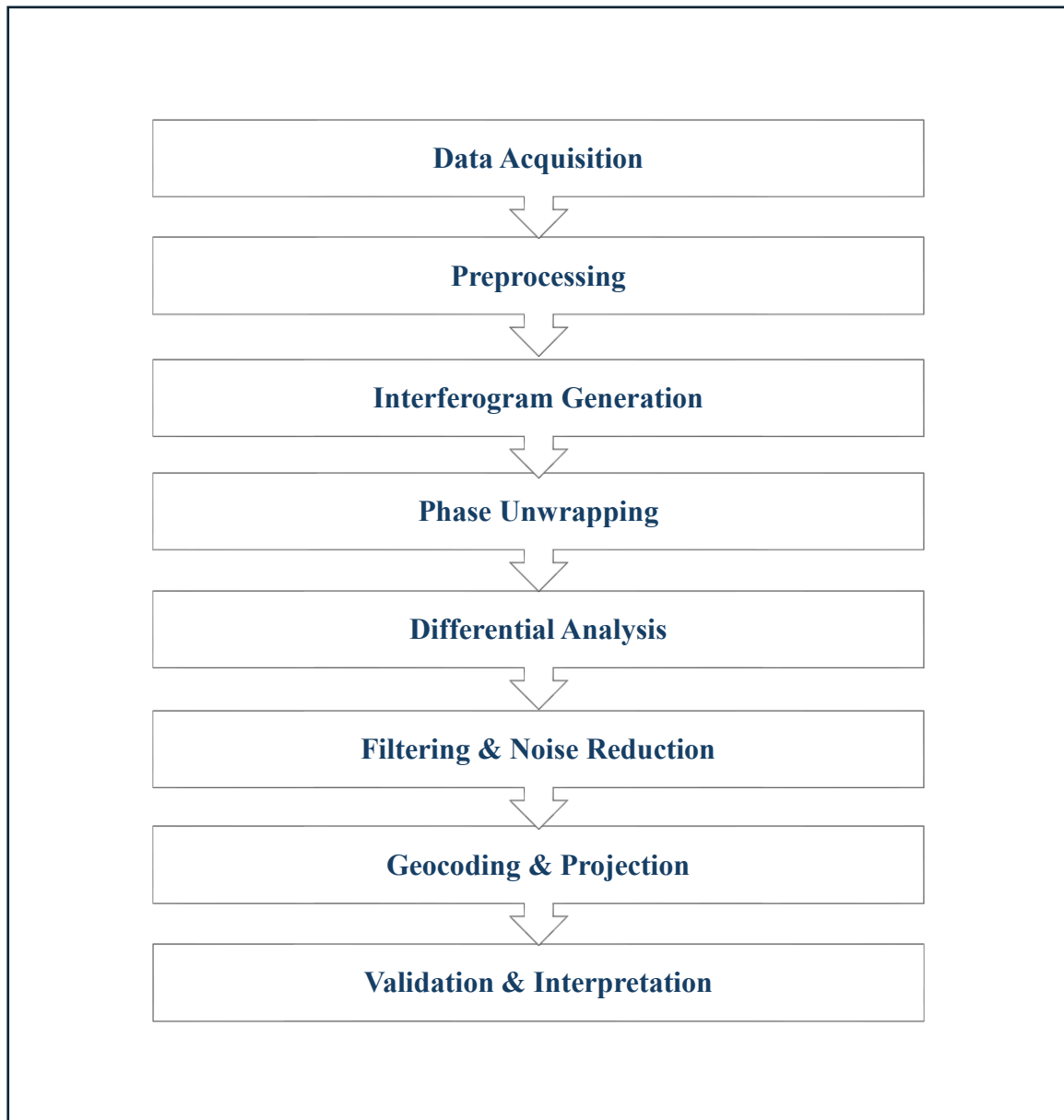
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
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S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 903/2

Bridge Chainage – 950+820

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

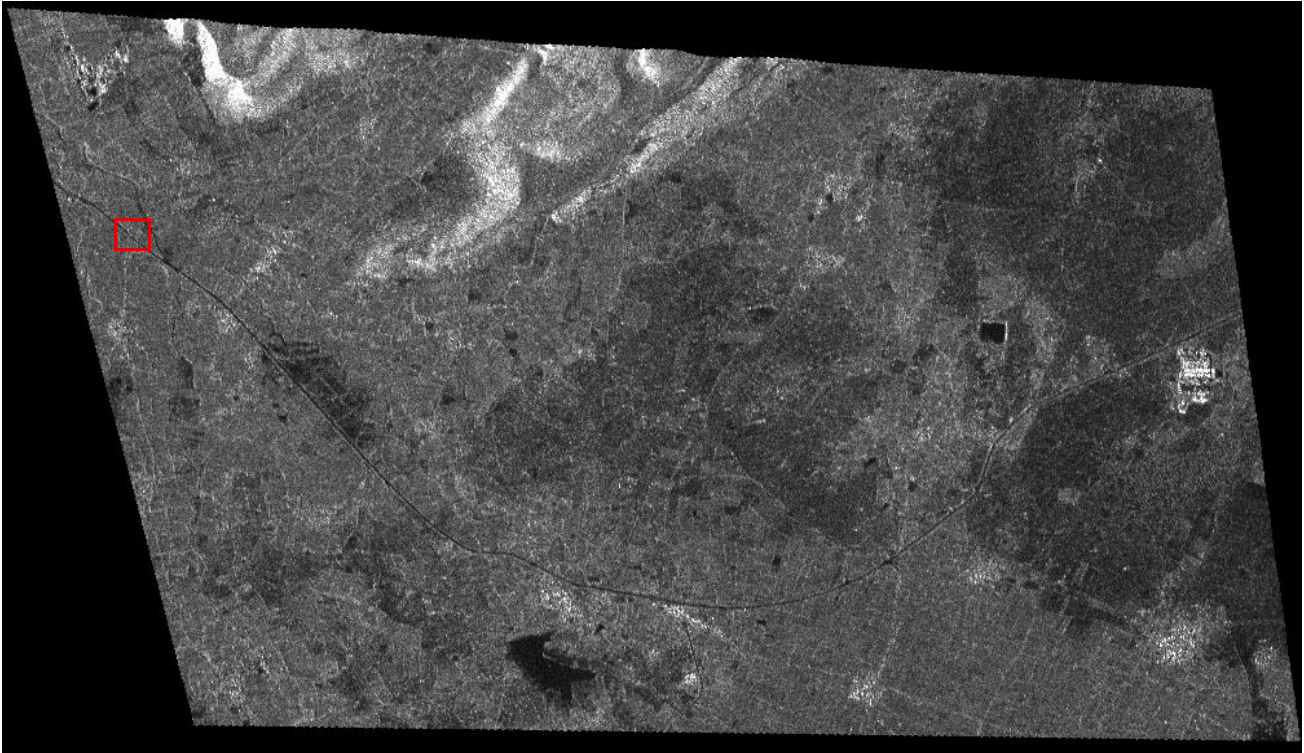
Flight Direction - DESCENDING

Polarization – VV+VH

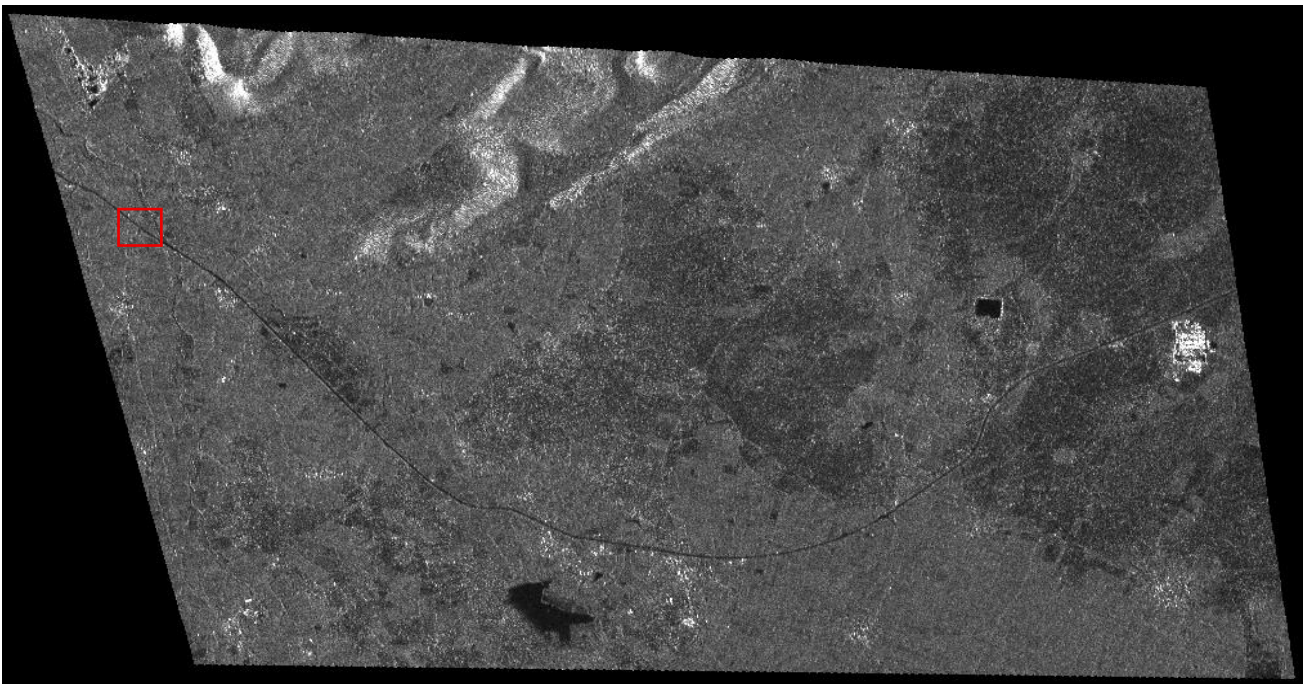
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

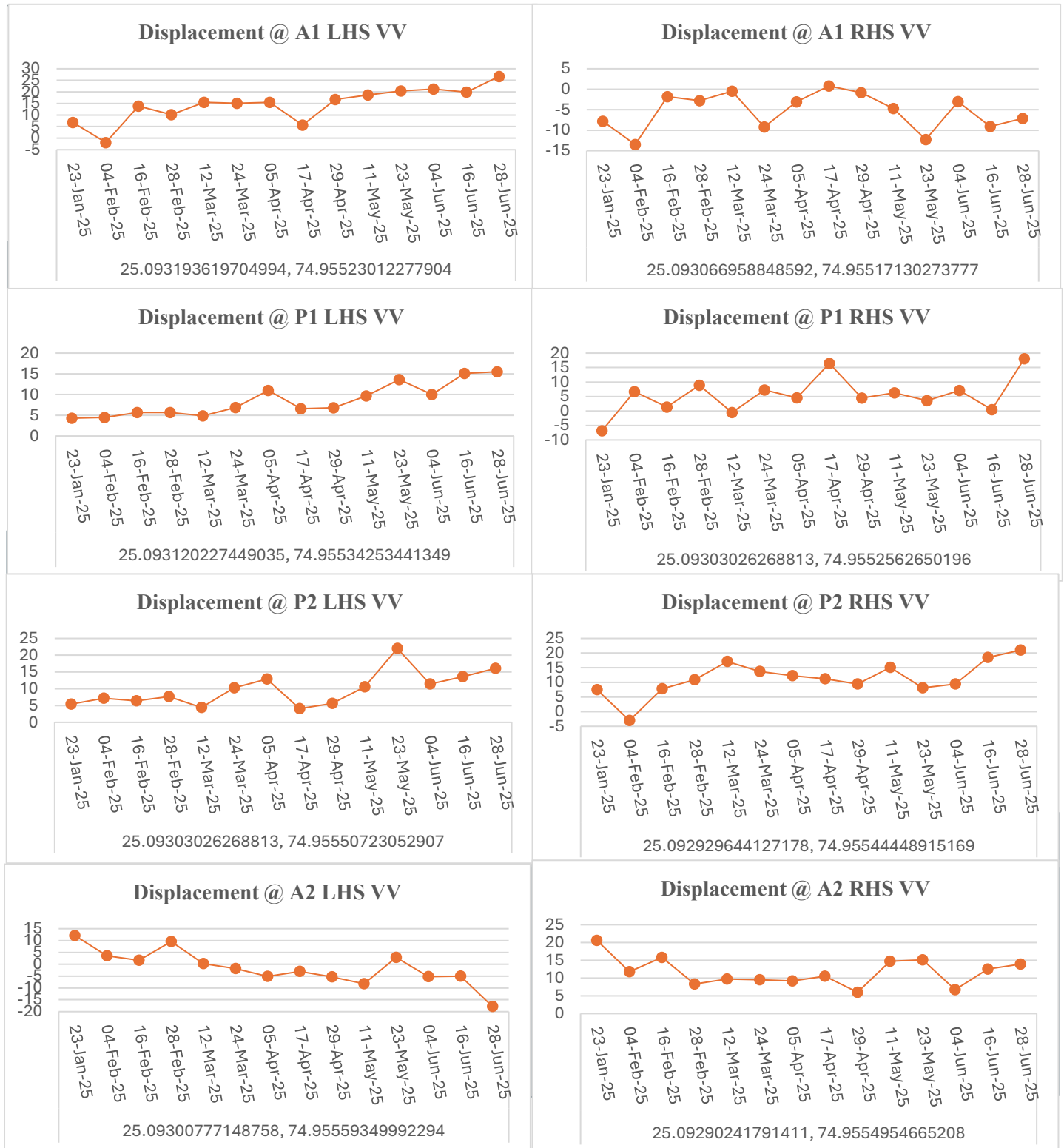
MNB – CH-950+820

Length – 51,72M



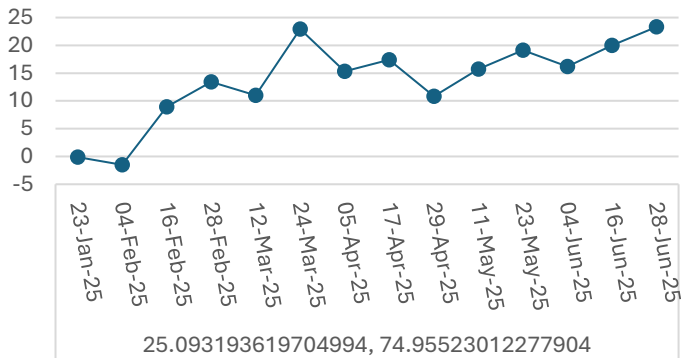
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
P1	Pier 1		
P2	Pier 2		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025

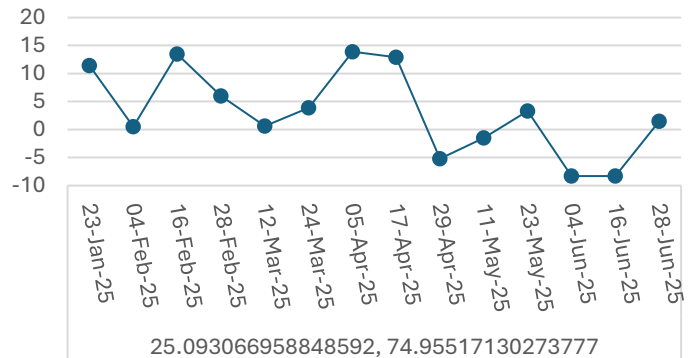


3.2 VH Displacement From 23 Jan 2025 To 28 June 2025

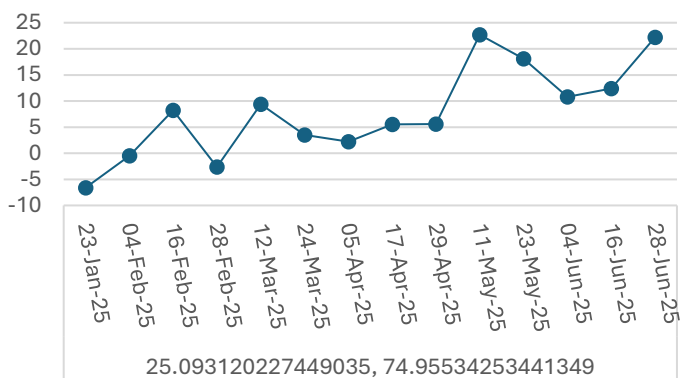
Displacement @ A1 LHS VH



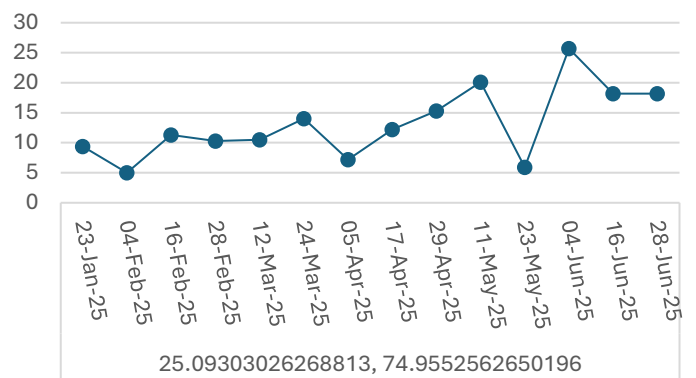
Displacement @ A1 RHS VH



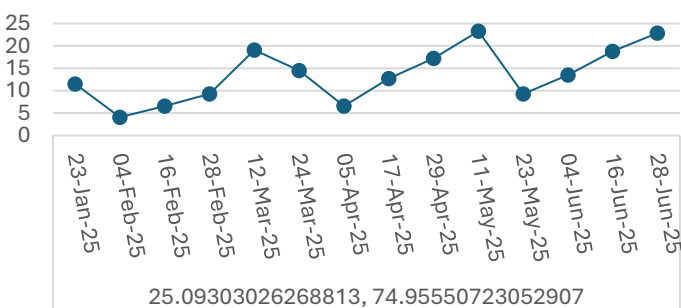
Displacement @ P1 LHS VH



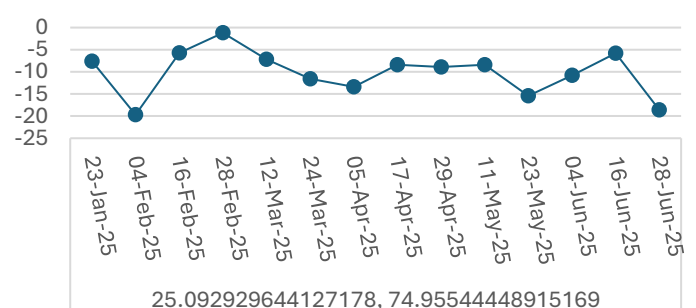
Displacement @ P1 RHS VH



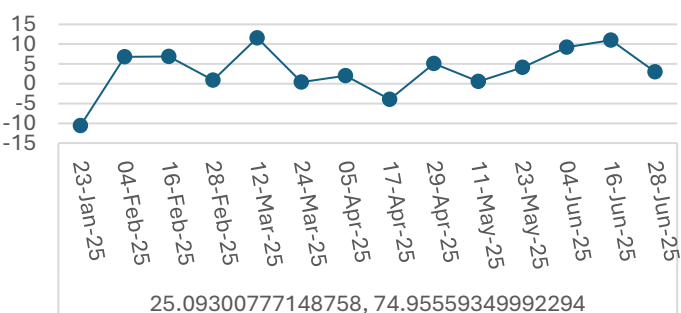
Displacement @ P2 LHS VH



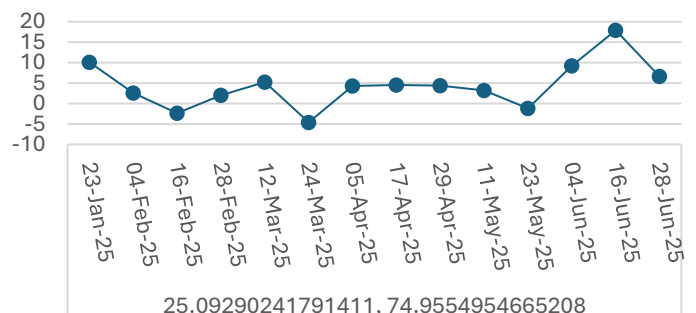
Displacement @ P2 RHS VH



Displacement @ A2 LHS VH



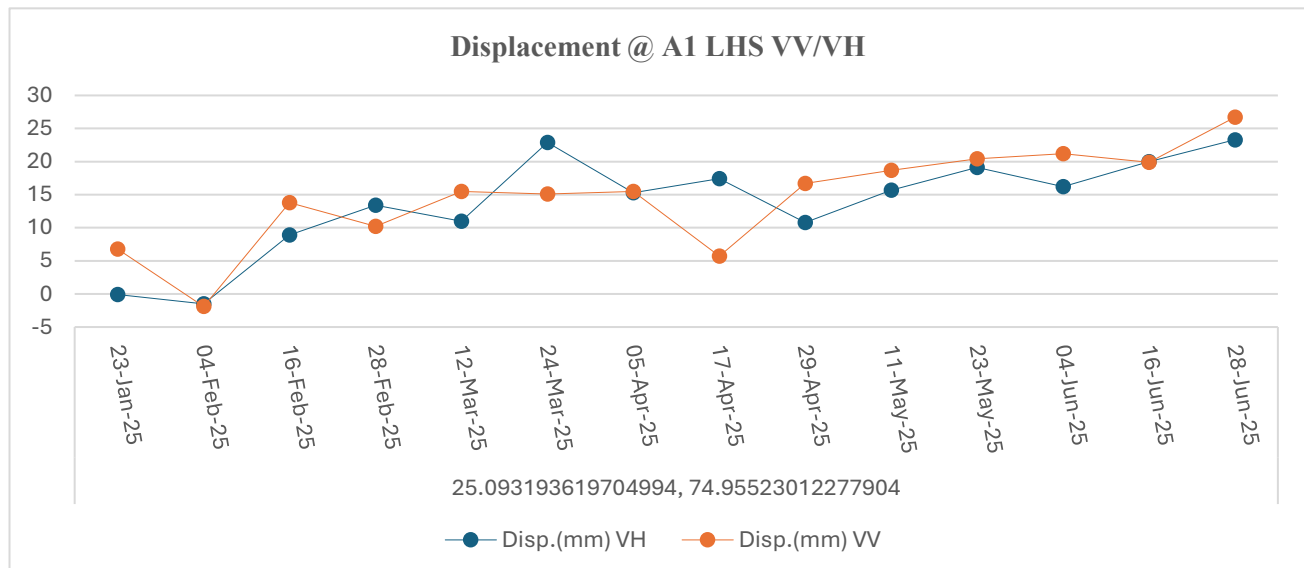
Displacement @ A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.093193619704994, 74.95523012277904)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** generally fluctuates within -10 mm to +10 mm through most of the monitoring period.
- **Horizontal displacement (VH)** shows stronger variability than VV, mostly ranging between -20 mm and 0 mm.

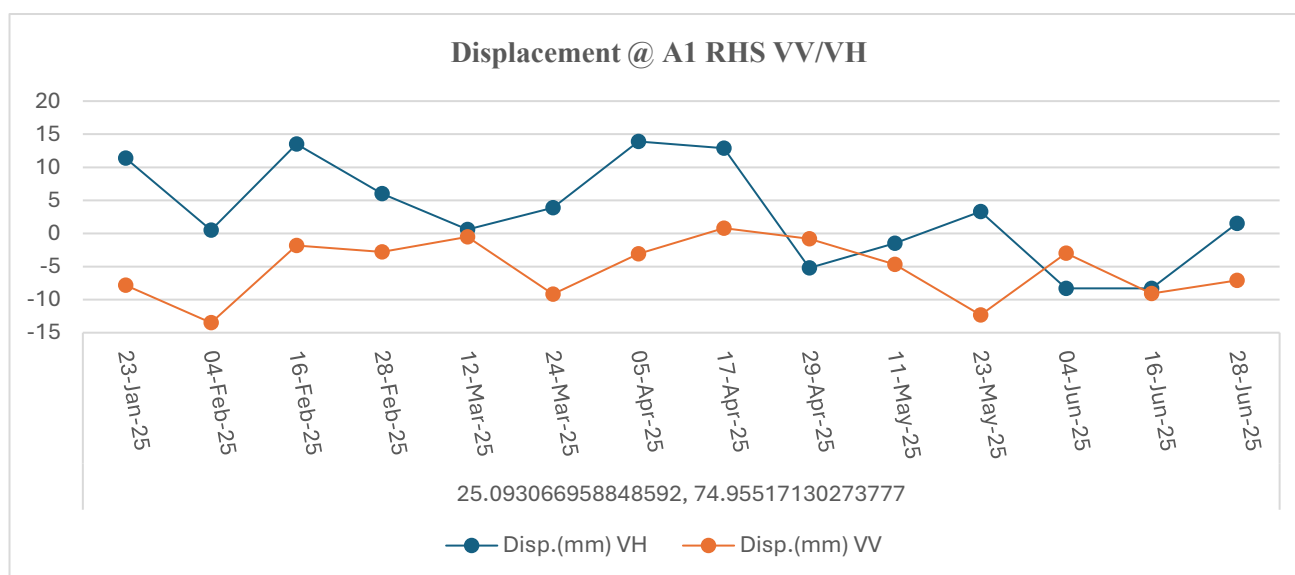
Critical Events & Anomalies:

- **August 2024:** A sharp negative spike in horizontal displacement (VH) reaches approximately -12 mm.
- **October & December 2024:** Two pronounced dips occur, with VH reaching -15 mm in October and nearly -20 mm in early December
- **March 2025:** Vertical displacement (VV) rises sharply to around +18 to +20 mm, representing the largest upward movement recorded for this location and indicating a sudden vertical deformation event.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.093066958848592, 74.95517130273777)



Key Observations:

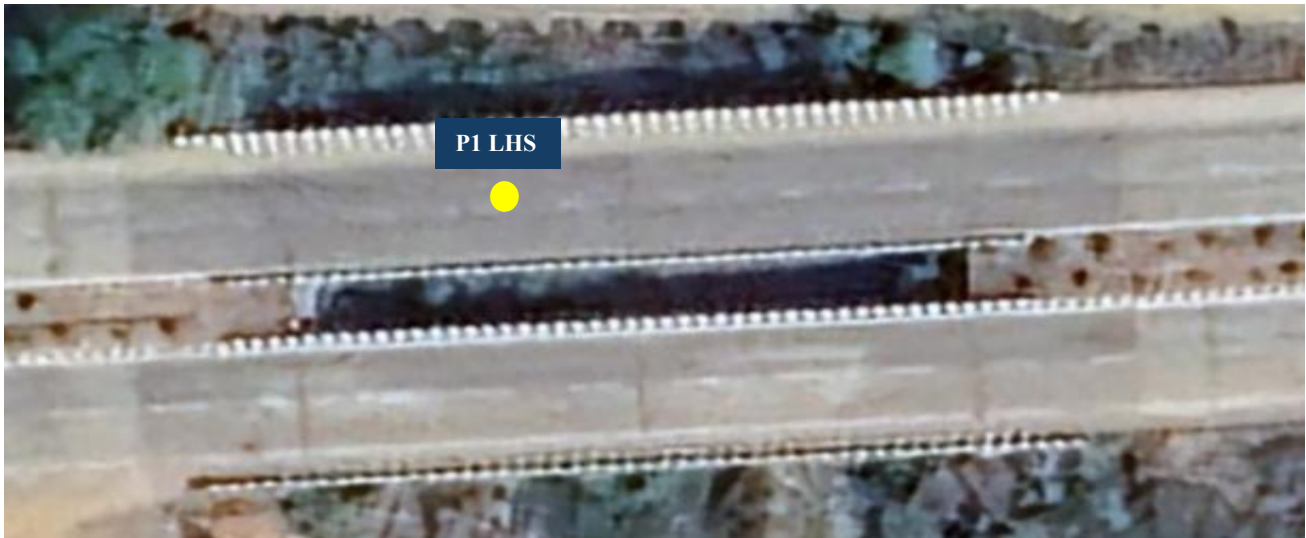
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH shows greater variability than VV, ranging from around -10 mm to +14 mm. Early measurements indicate positive horizontal movement, followed by alternating expansion and contraction phases toward the later part of the timeline.
- **Vertical displacement (VV)** predominantly remains in the negative range, fluctuating between approximately -15 mm and +2 mm throughout the monitoring period.

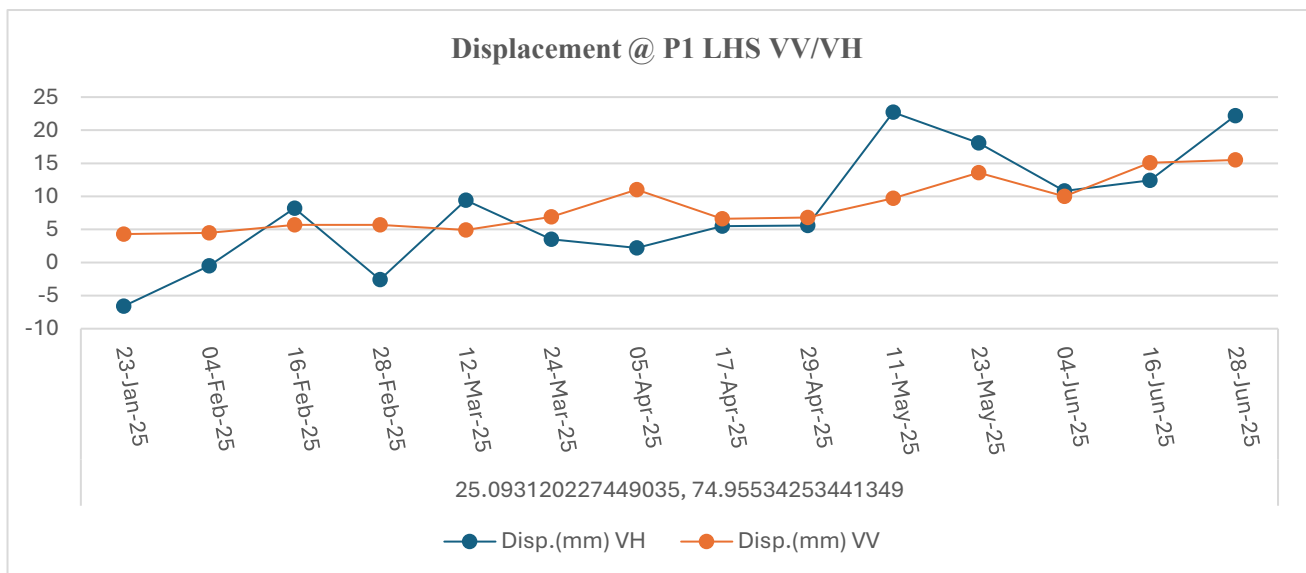
Critical Events & Anomalies:

- **Feb 2025:** A sharp vertical dip is observed on 04-Feb-25, where VV reaches approximately -14 mm, marking one of the largest downward vertical displacements in the dataset.
- **June 2025:** VH fluctuates between -8 mm and +2 mm, showing reduced horizontal stability. VV remains negative but shows slight recovery toward the end of June, stabilizing around -6 to -7 mm.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (25.093120227449035, 74.95534253441349)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** exhibits higher variability compared to VV, ranging from approximately -7 mm to $+22$ mm. Initial fluctuations include both negative and positive values.
- **Vertical displacement (VV)** shows a clear increasing trend over the monitoring period. Early values remain relatively stable between $+4$ mm and $+6$ mm, followed by a steady rise from late March onward.

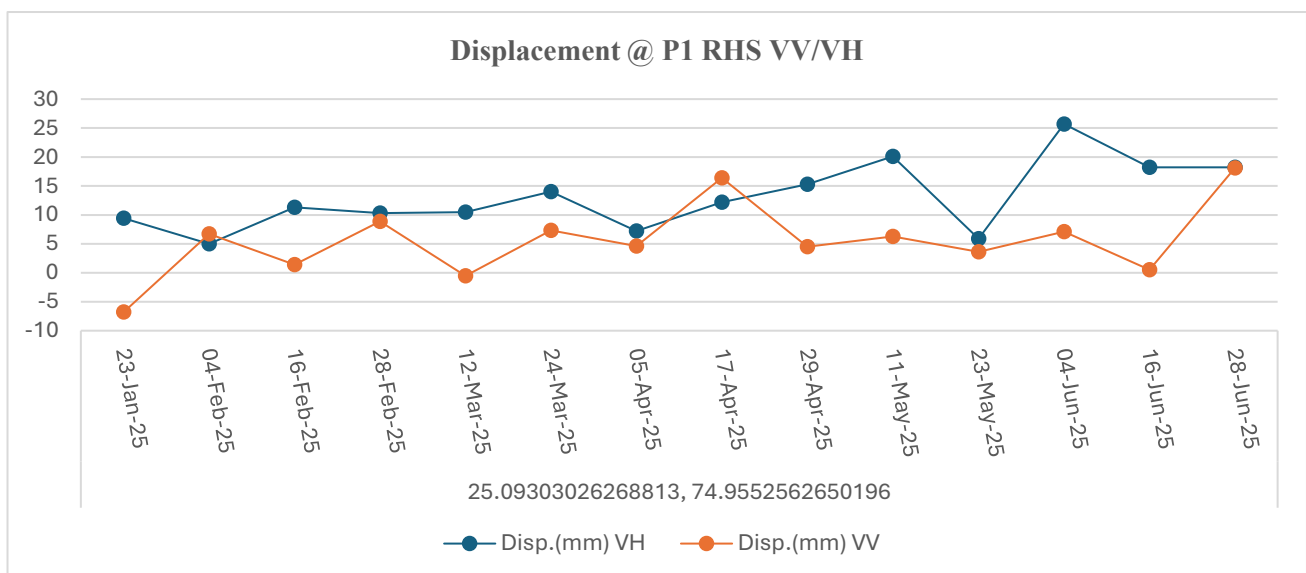
Critical Events & Anomalies:

- **January 2025:** A notable negative horizontal displacement occurs on 23-Jan-25, where VH drops to around -7 mm, marking the lowest horizontal value recorded at this point.
- **April 2025:** VV increases sharply to around $+11$ mm on 05-Apr-25, representing a significant vertical uplift event compared to earlier measurements.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (25.09303026268813, 74.9552562650196)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** demonstrates strong and sustained variability, generally remaining in the positive range and varying from about +5 mm to +25 mm.
- **Vertical displacement (VV)** shows moderate variability, fluctuating between approximately -7 mm and +18 mm over the monitoring period.

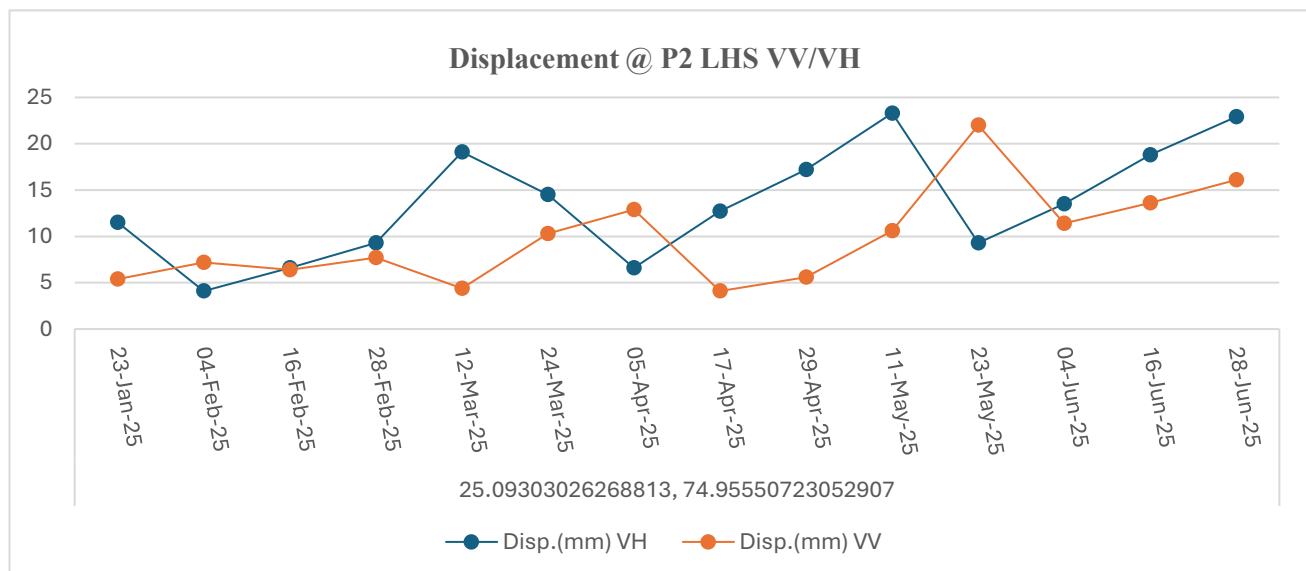
Critical Events & Anomalies:

- **January 2025:** VV records a notable negative value around -7 mm on 23-Jan-25, representing the largest downward vertical displacement in the early monitoring phase.
- **April 2025:** A pronounced vertical uplift is observed on Apr-25, where VV peaks near +16 mm, marking the highest vertical displacement until that point.
- **June 2025:** VV shows a strong recovery, increasing to approximately +18 mm on 28-Jun-25, while VH stabilizes near +18 mm, indicating sustained deformation in both components.

3.7 SAR Based Analysis at Pier 2 LHS:



P2 LHS (25.09303026268813, 74.95550723052907)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement VH** shows significant variability, ranging from about +4 mm to +23 mm. The horizontal component displays alternating increases and decreases, with an overall upward tendency toward the end of the monitoring period.
- **Vertical displacement (VV)** exhibits moderate to high variability, generally remaining in the positive range between approximately +4 mm and +22 mm.

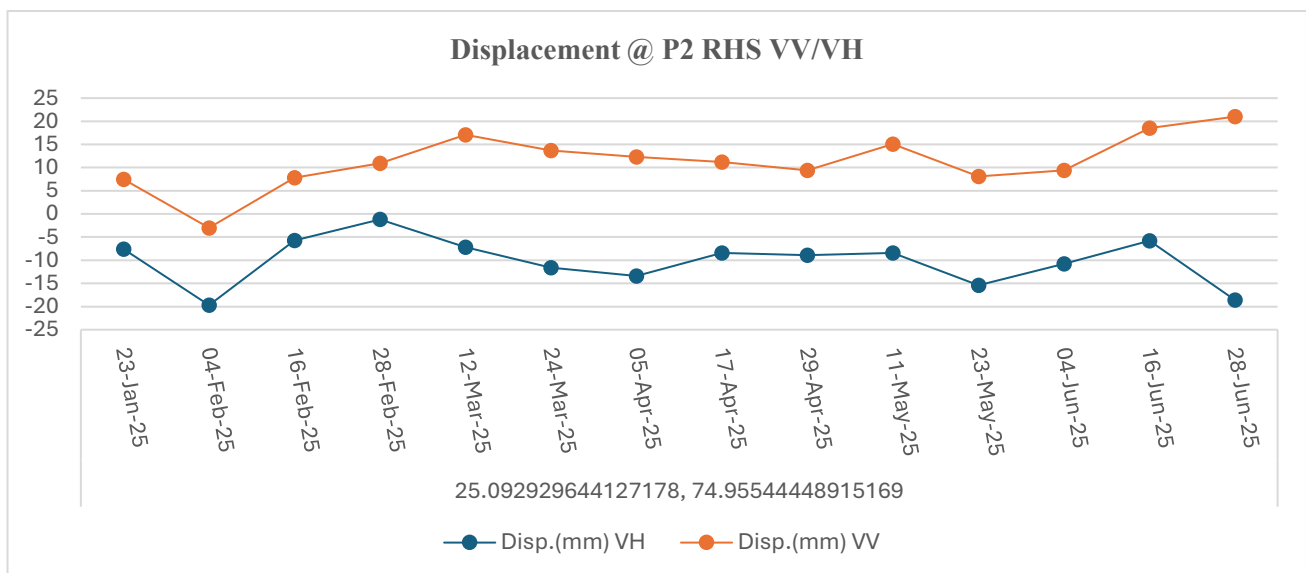
Critical Events & Anomalies:

- **February 2025:** VH records a local minimum around +4 mm on 04-Feb-25, representing the lowest horizontal displacement observed at this location during the early monitoring phase.
- **June 2025:** Both VV and VH show renewed increases. By Jun-25, VH rises to approximately +23 mm and VV stabilizes around +16 mm, indicating sustained deformation toward the end of the monitoring period.

3.8 SAR Based Analysis at Pier 2 RHS:



P2 RHS (25.092929644127178, 74.95544448915169)



Key Observations:

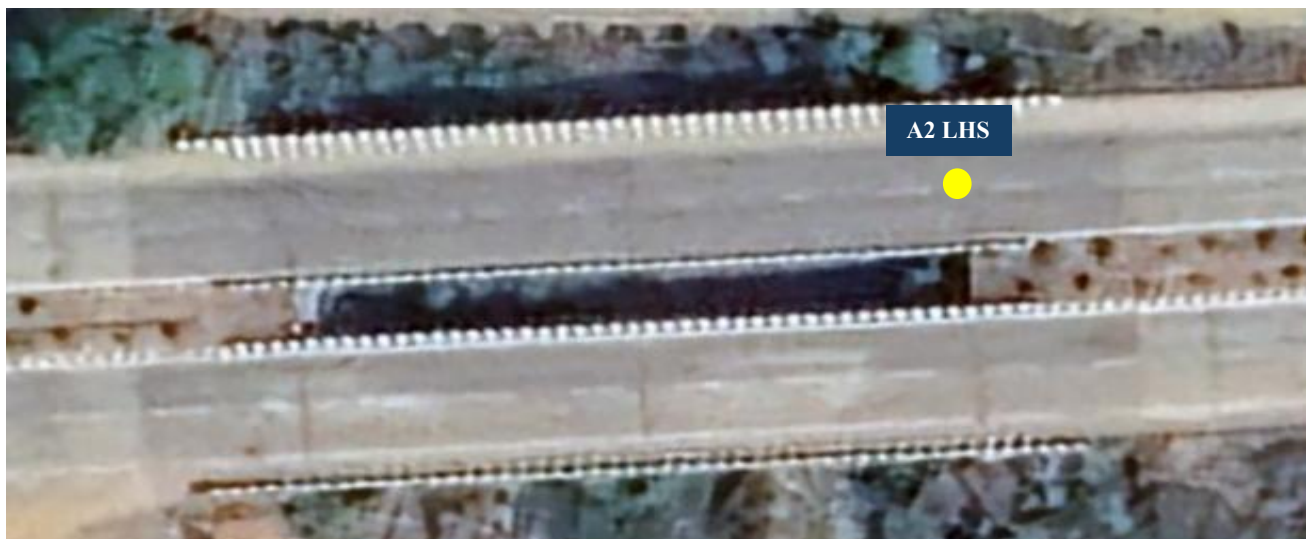
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH remains consistently negative, indicating sustained horizontal contraction. Values fluctuate between approximately -5 mm and -20 mm, with repeated negative peaks suggesting ongoing lateral deformation at this location.
- **Vertical displacement (VV)** shows a strong and consistent positive trend, remaining mostly in the uplift range between approximately +7 mm and +21 mm.

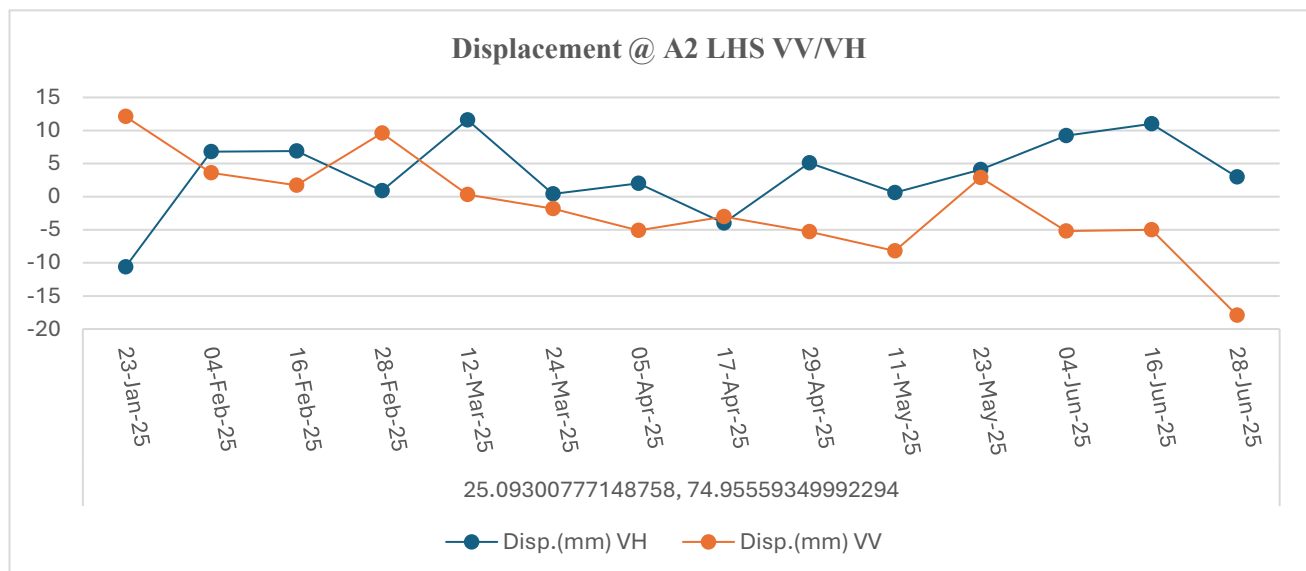
Critical Events & Anomalies:

- **February 2025:** VH shows partial recovery to around -2 mm on 28-Feb-25, representing the least negative horizontal displacement during the monitoring period.
- **June 2025:** VV reaches its highest recorded value (+21 mm on 28-Jun-25), indicating continued vertical uplift.

3.9 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.09300777148758, 74.95559349992294)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH):** VH fluctuates between approximately -10 mm and +11 mm, showing alternating contraction and expansion phases.
- **Vertical displacement (VV):** VV shows large variability with an overall downward trend. Values range from approximately +12 mm to -18 mm, transitioning from positive uplift in January–February to sustained negative displacement from April onward.

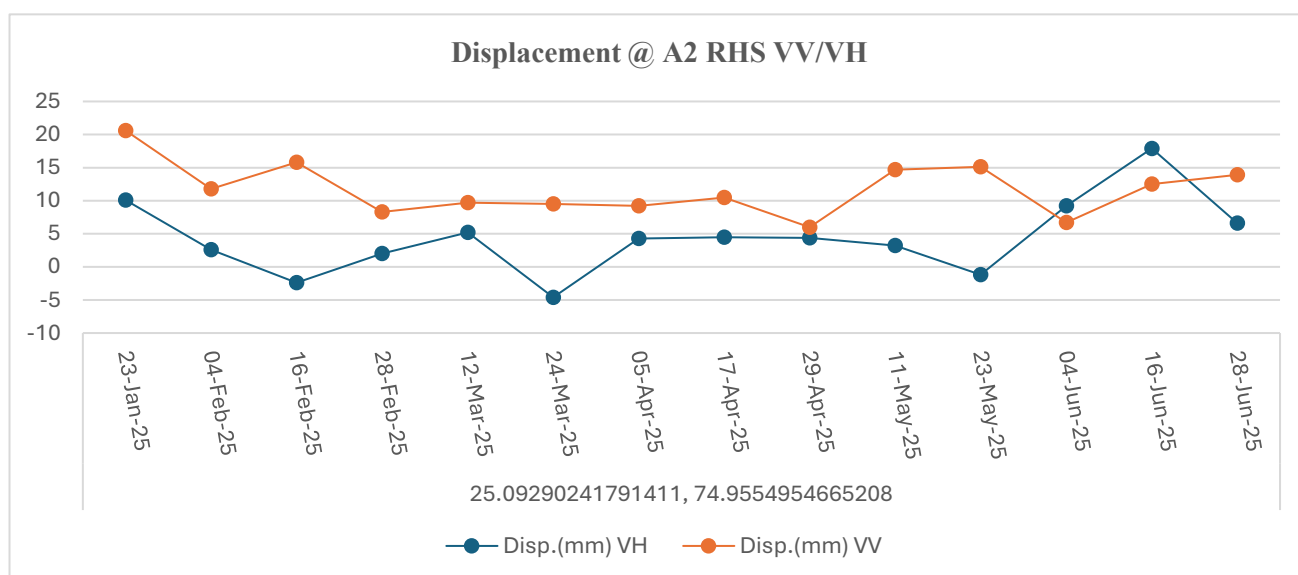
Critical Events & Anomalies:

- **February 2024:** VH stabilizes around +6 to +7 mm (04-Feb to 16-Feb), while VV declines from about +4 mm to +2 mm, indicating a shift from vertical uplift toward settlement.
- **June 2025:** VV experiences a sharp drop to approximately -18 mm on 28-Jun-25, representing the largest vertical settlement event in the dataset.

3.10 SAR Based Analysis at Abutment 2 LHS:



A2 RHS (25.09290241791411, 74.9554954665208)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH):** shows moderate variability, fluctuating between approximately –5 mm and +18 mm. The horizontal component alternates between contraction and expansion, with stronger positive movements observed toward the later monitoring stages.
- **Vertical displacement (VV):** remains consistently positive, indicating dominant vertical uplift throughout the monitoring period. Values range approximately from +6 mm to +20 mm.

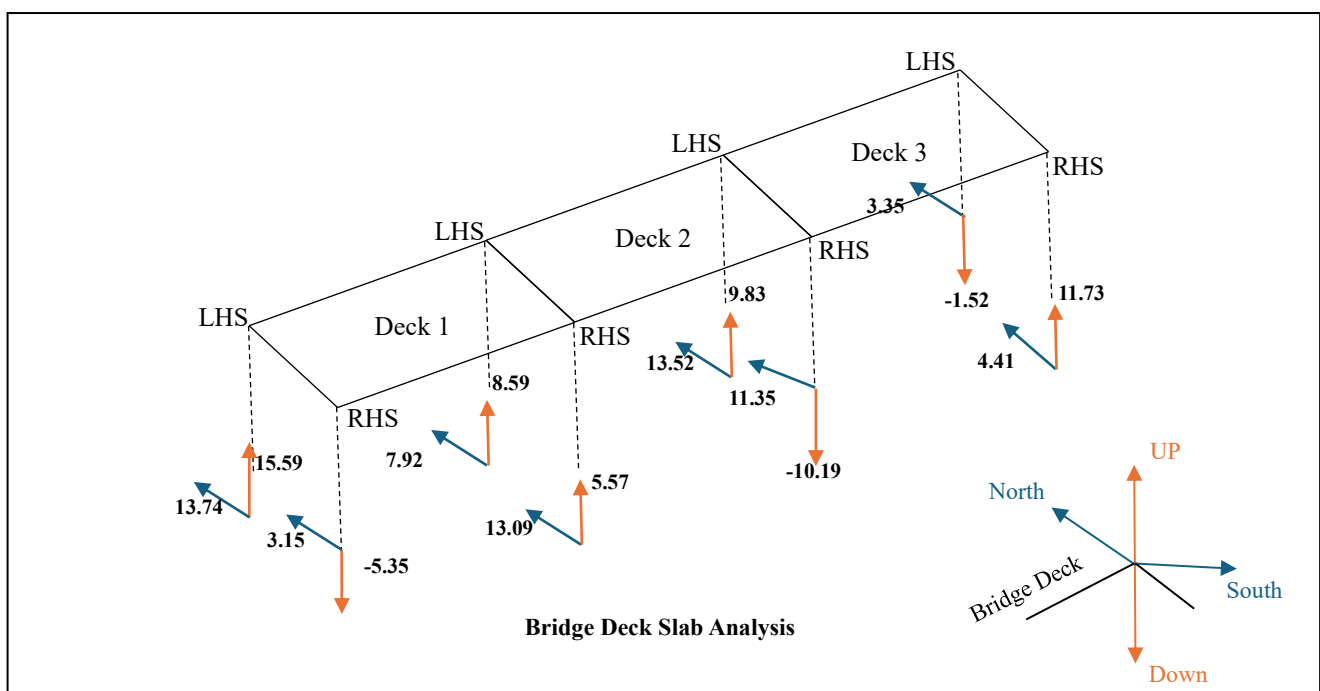
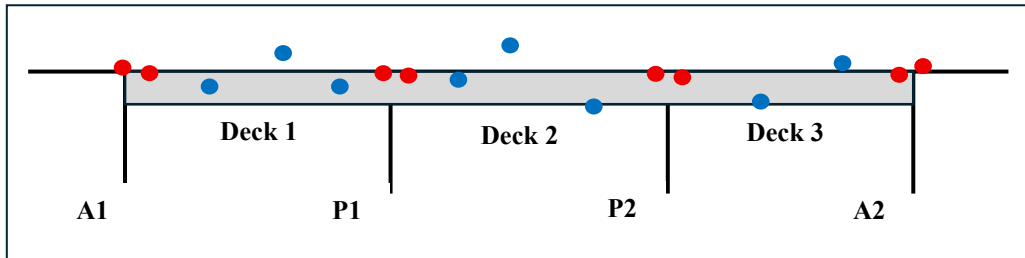
Critical Events & Anomalies:

- **January 2025:** VV records a high uplift value of around +20 mm on 23-Jan-25, representing the largest early vertical displacement at this location.
- **June 2025 :** VH decreases to about +6 mm on 28-Jun-25, while VV remains high near +14 mm, suggesting stabilization of horizontal movement with sustained vertical uplift.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- Vertical movements across Decks 1, 2, and 3 range from approximately -10.19 mm to $+15.59$ mm, indicating mild to moderate vertical response under operational and environmental loading.

Horizontal Displacement Behaviour:

- Deck 1 shows moderate lateral movement, with horizontal displacements of about 3.15 mm at the RHS and 13.74 mm at the LHS, indicating controlled lateral response. Deck 2 exhibits relatively higher horizontal activity, with displacements reaching up to 13.52 mm, suggesting increased lateral flexibility or load influence at this location. Deck 3 records lower horizontal movements, generally between 3.35 mm and 4.41 mm, reflecting reduced lateral demand toward the outer deck region.

Conclusion:

- The displacement behaviour observed across the bridge decks indicates that the structure is currently performing within acceptable deformation limits. While localized peaks in both vertical and horizontal displacement are present, the overall response remains stable, non-progressive, and structurally tolerable.